Conserve mechanical energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Track kinetic and potential energy when nonconservative work is negligible.

SUCCESS CRITERIA - I can...

1. Identify the system and what your energy calculation treats as true.
2. Keep the total energy consistent between positions.
3. Explain how height and speed trade off.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “mechanical energy” and “conserved” to explain your answer. Sentence starters are optional.

mechanical energy: total kinetic and potential energy in a model

conserved: unchanged in total under the stated conditions

system: the object or objects chosen for analysis

01 Warm-up: bring back an earlier idea

Lift a **3 kg** object by **5 m**.
1. Use g=**10 N/kg**.
2. **Find the change** in gravitational potential energy.

Word help: potential energy = energy associated with the configuration of interacting objects

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $K_i + U_i = K_f + U_f$; $K = \frac{1}{2}mv^2$; $U = mgh$.
Mechanical energy conserved when only conservative interactions transfer energy. Use g = **10 N/kg** near the surface.

Worked reasoning: A **1 kg** object drops **5 m** from rest with g=**10 m/s²** and no energy loss. It loses **50 J** of potential energy and gains **50 J** of kinetic energy, giving speed **10 m/s**.

Your turn: 1. **Draw energy bars** at the start and just before arrival.
2. **Label the system** and neglected losses.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Initial **Transfers** **Final**

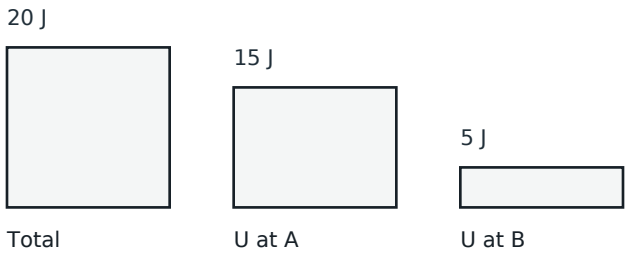
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A system has total mechanical energy **20 J**. At one point U=**15 J**; at another U=**5 J**. **Find K** at each point.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

04 Apply the physics

A **1 kg** ball is released from rest **2 m** above the floor. Use g = **10 N/kg**; ignore air resistance. **Complete an energy account** from release to just before reaching the floor. **Find initial U** and final K.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A cart loses mechanical energy on a rough track. Does total energy conservation fail? **Name the destination** to include.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

An impactor has **4** times the original mass at the same speed and stopping time. **Compare momentum** and average stopping-force size.
Use before and after snapshots to organize the momentum account.

Word help: momentum = mass times velocity, **including direction**

Sentence starter: Both … . They differ in … because … .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

An ideal system has **30 J** total. At a point U=**12 J**, what is K?
1. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.
2. Use g = **10 N/kg** when needed.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both … . They differ in … because … .

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Two frictionless ramps start at the same height and end at the same lower height.
1. **Compare final speeds** and **explain what need not be** equal.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.
3. Use g = **10 N/kg** when needed.

Sentence starter: Both … . They differ in … because … .

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3